

Seek Wander

Curated Luxury Journeys

Document Version	1.0 — Phase 3 Complete
Date	March 2026
Status	Production — Live on Vercel
Prepared by	Engineering & Product
Classification	Confidential

This document defines the complete product requirements for Seek Wander — a luxury AI travel concierge. It covers product vision, user personas, feature specifications, business model, and success metrics. All features described herein are implemented and live in production.

Executive Summary

Seek Wander is a premium web application that transforms how discerning travellers plan their journeys. Users describe their travel preferences through an intelligent 7-field intake form; within 15–20 seconds, the system returns a fully-verified, editorial-quality multi-day itinerary — complete with GPS-accurate activities, Michelin-calibre restaurant recommendations, curated hotel stays, real transit estimates, and destination photography.

The product is built on a **zero-hallucination data model**: every location generated by the AI is immediately enriched and verified via Google Places API before being shown to the user. Ratings, opening hours, and photos are ground-truthed in real time. This is not a generic travel app — it is a **digital concierge** for the luxury travel market.

Generation time	< 20 seconds	15–20s (p50)
Data accuracy	100% real places	Google Places verified
Cost per generation	< \$0.15	\$0.05–\$0.14 (60% cache)
Mobile experience	Full PWA	Installable, offline-capable
Uptime	99.9%	Vercel Edge (SLA)

Product Vision & Positioning

Vision: The world's most elegant travel planning experience — where every interaction feels like speaking with a seasoned, well-travelled concierge at a five-star hotel.

Positioning: Vogue meets National Geographic. The interface is intentionally editorial — warm paper tones, Cormorant Garamond serif headings, generous whitespace, and grayscale-to-colour photography transitions. This is not positioned as an 'AI tool' — the word 'AI' does not appear anywhere in the user-facing product. The technology is the invisible engine; the luxury outcome is the headline.

Brand Pillars

Editorial Aesthetic	Every screen feels like a luxury magazine spread. No border-radius. No clutter. Paper tones, ink blacks, burnt-orange accents.
Zero Hallucination	AI generates the itinerary structure; Google Places verifies every location. Users only see real, operational places with real photos and ratings.
Invisible Technology	The product outcome — a curated journey — is the brand. The underlying AI model is never referenced in user-facing copy.
Concierge Velocity	From preferences to complete itinerary in under 20 seconds. The speed of technology with the quality of a human expert.
Offline Resilience	Saved itineraries are accessible without connectivity. The app installs as a PWA and caches trips locally for travel use.

Target Audience

The Affluent Explorer

Demographics	35–55, HHI \$200k+, frequent international traveller
Pain Points	Existing tools (TripAdvisor, Google Maps) are generic and overwhelming. Travel agents are slow and impersonal. ChatGPT produces geographically inaccurate results.
Core Need	A curated, verified, beautiful itinerary that respects their taste — fast, and without requiring research expertise.

The Detail-Oriented Planner

Demographics	28–45, urban professional, values experience over possession
Pain Points	Spends hours researching and cross-referencing before every trip. Anxious about missing hidden gems or making suboptimal choices.
Core Need	A single trusted source that surfaces the 1% of places that matter — restaurants that require reservations, opening hours that are current, transit times that are realistic.

The Luxury Couple / Honeymoon Planner

Demographics	28–40, planning once-in-a-lifetime trips, budget-agnostic
Pain Points	Every result feels 'touristy'. Hard to find experiences that feel personal and exclusive. PDF itineraries from travel agencies are expensive and impersonal.
Core Need	A premium, shareable itinerary document they can export as a beautiful PDF dossier and send to their partner.

Core Feature Specifications

4.1 — 7-Field Concierge Intake Form

Staged inline expansion — each stage unlocks only after the previous is completed, creating a progressive disclosure pattern that mirrors a concierge conversation. No page reloads. All validation is instant.

1	Destination	Google Places Autocomplete	GPS-accurate lat/lng captured
2	Dates	Dual date picker	Departure + Return; duration badge computed
3	Travel Party	Pill selector	Solo / Couple / Family / Group
4	Pace	Card selector	Relaxed (3–4/day) / Moderate / Packed (6–7/day)
5	Budget Tier	Card selector	Premium \$\$ / Luxury \$\$\$ / Ultra-Luxury \$\$\$\$
6	Dietary	Multi-select	7 options incl. Halal, Kosher, Gluten-Free
7	Accommodation	Toggle + optional text	Needed (AI recommends) / Booked (user provides)

4.2 — AI Itinerary Generation Pipeline

Step 1: Validation	Zod schema parsing of all 12 request fields. 400 on failure.
Step 2: Rate Limit	Upstash Redis sliding window — 5 generations per user per hour.
Step 3: AI Generation	claude-sonnet-4-6 with injection-resistant system prompt. Returns chronological timeline[] per day with real place names and GPS coordinates.
Step 4: JSON Repair	sanitizeJson() regex sweep + fallback Claude repair call if JSON.parse() fails. Prevents malformed output from reaching users.
Step 5: Place Enrichment	enrichPlace() per timeline item — PlaceCache check first (30-day TTL), then Google Places Text Search + Place Details. Adds: photoUrl, rating, openNow, hoursOpen, priceLevel.
Step 6: Transit	Haversine formula — pure local math, zero API calls. Walking (5 km/h) and driving (25 km/h) estimates between consecutive stops.
Step 7: Cost Logging	CostLog Prisma record written (awaited) before response. Tracks AI tokens, Google API calls, cache hit/miss ratio.

4.3 — Split-Screen Itinerary Display

Desktop: 55% scrollable editorial timeline (left) + 45% interactive Google Map (right). Mobile: full-width timeline with collapsible map banner. Tabbed day navigation with AnimatePresence transitions.

- TimelineCard: unified card for activities AND meals — photo, name, description, rating, open/closed badge, price level, transit connector
- StayCard: Booking.com affiliate card for recommended hotels (AID 4013143)
- Hidden Gem: 1 per day — places 95% of tourists never find
- Day-centric SVG map markers with colour-coded legend (5 palette slots)
- Polyline connecting all stops in chronological order

4.4 — Trip Archive (/trips)

- Clerk-authenticated — only the owning user sees their trips
- IDOR-protected — /trips/[id] enforces userId ownership on every fetch
- Offline-capable — useOfflineTrips hook caches archive in localStorage (seek_wander_archive)
- Offline banner shown when navigator.onLine is false or /api/trips fetch fails
- UnauthenticatedState 'Velvet Rope' — editorial gate instead of redirect, Clerk modal sign-in
- EmptyTripsState 'Inspiration Hub' — editorial empty state with 3 destination teasers

4.5 — PDF & Sharing Features

- PDF Export: window.print() — zero dependencies. Tailwind print: modifiers hide nav, map, tabs. All days rendered sequentially. Framer Motion opacity overridden in @media print.
- Branded dossier header in print: Seek Wander wordmark, destination, date — hidden on screen.
- Public Sharing: /shared/[id] — intentionally public, no auth. Dynamic OG metadata (title, editorial description, destination photo) for iMessage/Slack/WhatsApp.
- ShareButton: navigator.share() (mobile native) with clipboard fallback + Sonner toast.

Business Model

Revenue Streams

Affiliate — Hotels	Booking.com AID 4013143. Every recommended stay links to Booking.com search. Commission on completed bookings via affiliate program.	Live
Stripe Paywall (Phase 12)	\$4.99 per generation after free tier (1 free generation per account). Stripe Checkout — no credit card required to browse.	Planned
Brand Partnerships	Destination tourism boards, luxury hotel chains paying for featured placement in recommended stays and editorial content.	Future

Unit Economics

Anthropic claude-sonnet-4-6 input	\$3.00 / 1M tokens
Anthropic claude-sonnet-4-6 output	\$15.00 / 1M tokens
Google Places Text Search	\$0.032 / call
Google Place Details	\$0.017 / call
Typical generation (cache cold)	\$0.08 — \$0.14
Typical generation (60% cache hit)	\$0.05 — \$0.09
PlaceCache hit savings (per item)	\$0.049 (Text Search + Details)
Vercel Pro hosting	\$20 / month flat
Supabase Pro	\$25 / month flat
Upstash Redis	~\$0 (within free tier for current volume)

Cost Efficiency Strategy

- PlaceCache (Supabase) stores Google API results for 30 days — 60–80% cache hit rate after first weeks of operation
- Vercel Cron deletes stale cache entries daily (14-day TTL on cold entries)
- Distance Matrix API removed — replaced by pure Haversine math (zero API cost)
- `parseOpenNow()` computes open/closed status locally from cached `hoursOpen` string — zero API calls on cache hits
- Rate limiting (5/hr) caps maximum AI spend per user
- CostLog tracks every generation: `/admin/metrics` dashboard gives real-time margin visibility

UX & Design Specifications

Design System

paper	#F5F0E8	Primary backgrounds — warm off-white
paper-dark	#EDE8DC	Card backgrounds, subtle depth
ink	#0A0A0A	Primary text, borders
ink-light	#6B6B6B	Secondary text, captions
emerald	#059669	Active states, OPEN badge
burnt-orange	#C2410C	CTA buttons, active tab indicator, CLOSED badge

Non-Negotiable Component Rules

- NO border-radius anywhere — rounded-none on every component (one exception: MobileMenu hamburger button uses rounded-full as an explicit mobile affordance)
- Thin borders only — border border-black/5 or border-b border-ink/20
- Framer Motion on all sections — fade-in-up: initial={{ opacity:0, y:24 }} animate={{ opacity:1, y:0 }} transition={{ duration:0.6, ease:'easeOut' }}
- Images: grayscale by default, full colour on hover (transition-all duration-700)
- Typography: Cormorant Garamond italic for headings, DM Sans for body/micro-copy
- Micro-copy: UPPERCASE, tracking-widest, DM Sans Bold
- HTML entities always: ' " ... · — never raw special characters

Notification System (Sonner)

Sonner toast library provides a three-state generation lifecycle — all three calls share id: 'curate-task' so the same toast mutates in place (no toast stacking):

Loading	Consulting the concierge...	(spinner)	Fetch start
Success	Itinerary Prepared	Your bespoke journey is ready for review.	200 response
Error	Concierge Busy	Our desk is at capacity. Please try again in a moment.	Any error / 429
Save OK	Passport Updated	This journey has been saved to your archive.	Successful DB write
Save Err	Save Failed	Unable to save this journey. Please try again.	DB exception

Mobile & PWA Requirements

Responsive Breakpoints

- Mobile (< 768px): full-width single-column layout, map collapses to h-52 banner above timeline
- Desktop (≥ 768px): 55%/45% split-screen — timeline left, sticky map right
- Hamburger menu hidden at md+ (desktop nav takes over)
- Print: single-column, all days sequential, map/nav/buttons hidden

PWA Specification

Installable	manifest.ts — standalone display, Seek Wander name, brand icon
Offline trips	useOfflineTrips + localStorage cache (seek_wander_archive)
Photo caching	Servist CacheFirst — 30-day TTL, 100-entry cap on Google Photos URLs
iOS integration	appleWebApp meta — capable:true, statusBarStyle:default
Service Worker	src/app/sw.ts → public/sw.js (disabled in development)

Mobile Navigation Architecture

The mobile overlay uses `createPortal(overlay, document.body)` to escape the Navbar's CSS stacking context (created by `backdrop-blur-sm` on `motion.nav`). Without the portal, fixed `inset-0` anchors to the Navbar element (~64px) rather than the viewport — making the overlay appear transparent. This is a documented architectural rule for all future overlay components.

Security Requirements

Input Validation	Zod ItinerarySchema.safeParse()	All API POST bodies
Prompt Injection	SYSTEM_PROMPT as system: param (not in messages[])	AI generation
Rate Limiting	Upstash slidingWindow(5, '1 h') per userId/IP	POST /api/itinerary
AI Self-Healing	sanitizeJson() + JSON_REPAIR_PROMPT fallback Anthropic call	AI response parsing
IDOR Prevention	Post-fetch userId ownership check — same notFound() for both	/trips/[id]
API Key Isolation	Client key: HTTP referrer restricted. Server key: no referrer	Google Places
Auth Guard	Clerk auth() + userId check — returns 401 if absent	/api/trips
Admin Neutral 404	notFound() (not redirect) for non-admin /admin/metrics	Admin route
Cron Security	Authorization: Bearer CRON_SECRET header check	/api/cron/cleanup
HTTP Headers	X-Frame-Options DENY, nosniff, strict-origin Referrer-Policy	All routes

Success Metrics & KPIs

Product Metrics

Generation completion rate	% of started generations that return a result	> 95%
Generation latency (p50)	Time from form submit to itinerary display	< 20s
Cache hit rate	PlaceCache hits / total enrichment calls	> 60% (steady state)
Cost per generation	AI + Google API spend per itinerary	< \$0.12
Save rate	% of generated itineraries saved to archive	> 30%
Share rate	% of saved trips shared via ShareButton	> 15%
PDF export rate	% of saved trips exported to PDF	> 10%
Mobile install rate	PWA installs / mobile unique visitors	> 5%

Business Metrics

Affiliate click-through rate	Clicks on StayCard Booking.com links / itineraries with stays shown
Affiliate conversion rate	Booking.com confirmed bookings via AID 4013143 / affiliate clicks
Monthly affiliate revenue	Commission from Booking.com partner dashboard
Cost margin	(Revenue from affiliate + future Stripe) / Total AI + Google API cost
Monthly Active Users	Unique Clerk userIds with ≥ 1 generation in rolling 30 days

Phase Roadmap

1 — Core Engine	Next.js + Claude + Google Maps baseline	Complete
2 — Concierge UX	7-field intake form, split-screen results, Google enrichment, Haversine transit	Complete
3 — Ultra-Luxury UI	Tabbed day nav, cost badges, transit connectors, day-centric map markers	Complete
4 — Auth	Clerk v6 integration, conditional ClerkProvider, NavbarAuth, custom 404	Complete
5 — Persistence	Prisma + Supabase, saveTrip server action, /trips archive, /trips/[id], IDOR enforcement	Complete
6 — PWA & Sharing	@serwist/next, /shared/[id] OG metadata, ShareButton, PDF/print export	Complete
7 — Timeline Refactor	timeline: TimelineItem[] canonical shape, normalizeDayPlan() backward-compatible shim	Complete
8 — Margin Protection	PlaceCache, pure Haversine, CostLog, /admin/metrics, Vercel Cron, Zod, security headers	Complete
9 — Affiliate Monetization	Booking.com AID 4013143, StayCard, accommodation branching, recommendedStays[]	Complete
10 — UX & Polish	GenerationLoader, Sonner toasts, UnauthenticatedState, EmptyTripsState, MobileMenu portal fix	Complete
11 — Mobile & Resilience	useOfflineTrips, /api/trips, Upstash rate limiting, AI JSON self-healing, dynamic OG metadata	Complete
12 — Stripe Paywall	\$4.99 per generation after 1 free — Stripe Checkout, usage tracking	Next
13 — Brand Partnerships	Featured placement API for destination tourism boards and luxury hotel chains	Future